Inheritance in C

Simulating Object-Oriented Inheritance via Struct Composition

Topic Overview & Technique

Object-oriented languages like C++ or Python have built-in keywords for inheritance. In C, however, struct composition achieves the exact same memory layout and behavior:

- **Base Struct as First Member:** By embedding the parent struct as the very first member of a child struct, C guarantees pointer compatibility.
- **Type Casting / Upcasting:** A pointer to a child struct can be safely cast or passed as a pointer to its parent struct.
- **Single, Hierarchical, & Multilevel:** Structs can be nested to model complex familial hierarchies (e.g., `GamingYouTuber` → `YouTuber` → `SocialMediaUser`).

Task Solutions

Task 1: Base Class – `SocialMediaUser`

```c
#include <stdio.h>
#include <string.h>

typedef struct {
    char username[50];
    int followers;
} SocialMediaUser;

void displayProfile(SocialMediaUser *user) {
    printf("Username: %s | Followers: %d\n", user->username, user->
       followers);
}

int main() {
    SocialMediaUser user1;
    strcpy(user1.username, "TechGamer");
    user1.followers = 15000;
    displayProfile(&user1);
    return 0;
}
```

Task 2: Single Inheritance – `YouTuber`

```c
typedef struct {
    SocialMediaUser base;   /* Inherited parent member */
    char channelName[50];   /* Child property */
} YouTuber;

void uploadVideo(YouTuber *yt, const char *title) {
```

```
    printf("Video [%s] uploaded to [%s]\n", title, yt->channelName);
}
```

Task 3 & 5: Hierarchical Inheritance – `Podcaster` & `InstagramInfluencer`

Multiple distinct child structs inheriting from the same base `SocialMediaUser` parent.

```
typedef struct {
    SocialMediaUser base;
    char podcastName[50];
} Podcaster;

typedef struct {
    SocialMediaUser base;
} InstagramInfluencer;
```

Task 4: Multilevel Inheritance – `GamingYouTuber`

Chaining inheritance levels where a child inherits from another child struct.

```
typedef struct {
    YouTuber youtuber;      /* Contains YouTuber -> contains
       SocialMediaUser */
} GamingYouTuber;

void streamGame(GamingYouTuber *gyt, const char *gameName) {
    printf("%s is now streaming %s on %s\n",
           gyt->youtuber.base.username, gameName, gyt->youtuber.channelName
               );
}
```

Summary of Inheritance Types

#	Class / Struct	Inheritance Type
1	`SocialMediaUser`	Base (parent) struct
2	`YouTuber`	**Single** inheritance
3	`Podcaster`	**Hierarchical** inheritance
4	`GamingYouTuber`	**Multilevel** inheritance
5	`InstagramInfluencer`	**Hierarchical** inheritance

Key Takeaway: Placing the parent struct as the first element of the child struct replicates the exact memory offset layout of object-oriented inheritance in standard C!